

KUSH PATEL

Pittsburgh, PA

📞 732-421-8945 ✉ kushpate@andrew.cmu.edu [in linkedin.com/in/patelkush1508](https://www.linkedin.com/in/patelkush1508) github.com/kushp15

Summary

Seeking 2026 AI/ML Engineering internship, combining technical expertise with strong collaboration and problem-solving skills to design scalable AI/ML solutions and drive business impact

Education

Carnegie Mellon University

May 2027

Masters of Science in Artificial Intelligence Engineering - Information Security

Pittsburgh, PA

Coursework: Machine Learning, Systems & Tool Chain in AI Engineering, Information Security, Cloud Security

Rutgers University, New Brunswick

Sept 2022 – Aug 2024

Bachelor of Science in Computer Science, Minor in Business Administration — GPA : 3.6/4.0

Piscataway, NJ

Coursework: Artificial Intelligence, Deep Learning, Computer Security, Operating systems & Design, Data Structure

Activities: RUMAD (Mentor), HinduYuva (Leadership), Hackathons (3, 1×Winner), Chess Club

Technical Skills

Programming Languages: Python (Advanced), Java, C, PostgreSQL, NoSQL, React

Frameworks & Libraries: PyTorch (Advanced), TensorFlow, NumPy, Pandas, OpenCV, Apache Kafka, Spark

Tools & Technologies: GitHub, Docker, Jenkins, Postman, RESTful APIs

Cloud & Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS), Windows, Linux

Experience

Advantest America, Inc

Jun 2024 – Present

Technology Specialist, Developing

South Plainfield, NJ

- Built and deployed an **AI Agent using LLMs and Model Context Protocol**, delivering real-time production reports that improved decision-making across global operations
- Designed distributed **ETL pipelines ingesting 150M+ records** via REST APIs into PostgreSQL, ensuring real-time sync and enterprise-grade reliability
- Engineered an ERP-integrated scheduling platform in Python/C++ that **cut planning time by 90%** and boosted operational efficiency
- Implemented **CI/CD workflows** with GitHub Actions, **reducing deployment time by 40%** while improving reliability through automated testing
- Integrated robotic systems with TCP/IP protocols, increasing PCB measurement **accuracy and speed by 22%**

Advantest America, Inc

Aug 2022 – May 2024

Technology Intern

South Plainfield, NJ

- Oversaw **1000+ users** Azure onboarding, email setup, and Intune deployments to boost efficiency and cut errors
- Maintained VMware, Windows Server, and Active Directory infrastructure to **ensure reliability and uptime**
- Led **audits and backups for 1,200+ devices** across global sites to strengthen data integrity and inventory tracking

Projects

Computer Vision PCB Inspection Tool | Python, EasyOCR, Linux

2025

- Created a real-time PCB inspection system to automate layer sequence validation using OpenCV and OCR
- Structured logging via **REST APIs to cut inspection latency by 22%**, result real-time data streams and analysis
- Optimized **image processing algorithms** to enhance inspection accuracy and throughput, **saving \$1M annually by reducing manufacturing defects**

Lazy Track | Python, Transformers, APIs, SQLite

2025

- Developed an **AI Agent** to monitor emails and extract job-related context using a **Model Context Protocol (MCP)**
- Automated status updates and reminders, improving job search **efficiency by over 70% with APIs**

Fraud Email Detector Model | Python, PyTorch, GCP, claude

2025

- Built a binary classification model using **TF-IDF and embeddings** to detect fraud in 100K+ real-world email samples
- Achieved 93% accuracy** and reduced false positives, improving inbox safety for end users and reducing phishing risk

Leadership / Extracurricular

BAPS Charities

Feb 2021 - Present

Mentor & Co-ordinator

Edison, NJ

- Led a team of 8, running weekly check-ins to coordinate project milestones, task ownership, and progress tracking
- Organized and facilitated technical workshops, improving team collaboration and knowledge sharing across projects